

Epistemic collapse in multi-agent systems

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September 2026 — working summary

Motivation

In the simulation framework of *Control Charts for Multi-agent Systems* (arXiv:2605.11135), N LLM agents each hold a vector memory, answer questions from a shared pool by retrieving from that memory, exchange answers with peers, and are instructed to say “I don’t know” (IDK) when retrieval provides no grounding. That paper studied an *exogenous* failure — adversarial injection — and asked whether statistical monitoring detects it. This project studies the *endogenous* failure: the regime in which an agent, or every agent, can answer with nothing but IDK. We call this **epistemic collapse**, and we want to know when it happens, whether it is a sharp threshold or a graded frontier, how fast it arrives, which factors drive it, and whether the two natural levels — *agent-level* (a given agent cannot answer) and *system-level* (no agent can answer) — move together or apart. Sitting behind all of this is a trade-off the static framework cannot express: in a changing world, an agent that retains evidence too long is *wrong*, and one that discards it fast enough to be right may fall below the threshold at which the collective can know anything at all.

Framework

Model. We work with a deliberately small model (toy simulator v2) that keeps the paper’s ask/answer/IDK mechanics but adds the variables the question needs. N agents face M questions; question q changes each step with probability λ_q (log-normal across questions, mean $\bar{\lambda}$, dispersion σ_λ ; static questions are the $\lambda_q \rightarrow 0$ limit). Every stored answer carries two timestamps: t_{obs} , when it was observed from the environment — *provenance travels unchanged through exchange* — and t_{recv} , when this agent received its copy; on receiving a duplicate, the newer t_{obs} wins. Each agent spends a budget of B actions per step: a fraction α observing the environment (question drawn from π_{env} , observation exact and freshly stamped) and the rest asking peers (question drawn by policy σ weighted by demand π_{ask} , peer drawn on contact graph G). An agent *answers* q only if its estimated probability of still being right, $e^{-\lambda_q(t-t_{\text{obs}})}$, is at least a caution level c — so evidence expires at age $\theta_q = \ln(1/c)/\lambda_q$ — and only if it received its copy within a memory lifetime τ .

Metrics. Agent-level collapse probability $P(\text{IDK}) = 1 - \text{knows over (agent, question) pairs}$; system-level collapse $P(\text{dead}) = \text{fraction of questions no agent can answer}$; the answers actually given split into correct vs. stale. All are means over the last 200 of 800 steps, 10 seeds.

Mean-field picture. Per question, knowledge is a

susceptible–infected–susceptible contagion: a copy is born when an ask reaches a holder, and dies when it expires. With bandwidth $m = (1 - \alpha)B$ this gives a reproduction number $R_0 \approx m\tau/M$, N -invariant to first order. Three timescales organise everything: memory τ , the per-question observation interval $M/(\alpha BN)$, and evidence validity θ_q . If $\tau = \infty$, communication cannot affect system-level survival at all — a lineage dies θ_q after observation no matter how much it circulates — and only spreads evidence farther (agent level). Communication becomes load-bearing at the system level exactly when $\tau < M/(\alpha BN) < \theta_q$: the observer forgets before anyone re-observes and before the evidence goes stale, so a lineage survives only by circulating — the SIS threshold confined to the validity window. A healthy system needs all three conditions at once: $m\tau/M \gtrsim 1$ (spread), θ_q large enough relative to re-observation (someone is watching in time), and caution matched to speed (don’t be stale). The candidate headline result is that the feasibility window closes entirely when the environment is fast enough relative to bandwidth — the epistemic analogue of the control-chart paper’s learning–security trade-off.

Experimental setup

Two standalone numpy simulators, both implementing the repo’s asking/answering rules. **v1** (receipt-clock expiry τ_{mem} , independent ask rate K and observation rate ε , no caution) exists to test the SIS prediction cleanly. **v2** is the model above. The published simulator itself has none of these levers — memory never expires (a control run confirmed it cannot produce system-level death at any pool size: without a death term, knowledge only accumulates), so the toy models carry the design until the repo is extended. The v2 design is a factor-structured sweep: variables sort into four factors — *agent*, *communication*, *environment*, *observation* (the environment→agent interface) — and each factor contributes one panel varying its intrinsic lever with two secondary variables (lightness / line-style), all other factors held at a common baseline chosen so the three clocks are ordered $\tau < M/(\alpha BN) < \theta$: $N=20$, $M=50$, $B=5$, $\alpha=0.01$ (99 asks per observation), $\tau=20$, $c=0.5$, $\bar{\lambda}=0.003$ ($\theta \approx 231$, $R_0 \approx 2$), flat π_{env} and π_{ask} , full mesh. At baseline the system is healthy: $P(\text{IDK}) = 0.14$, $P(\text{dead}) = 0.02$. The full grid is 1,380 runs (≈ 75 s on 30 cores), so iteration is effectively free.

Results

v1: collapse is an SIS threshold. With expiry on the receipt clock, the fraction of questions alive collapses as a

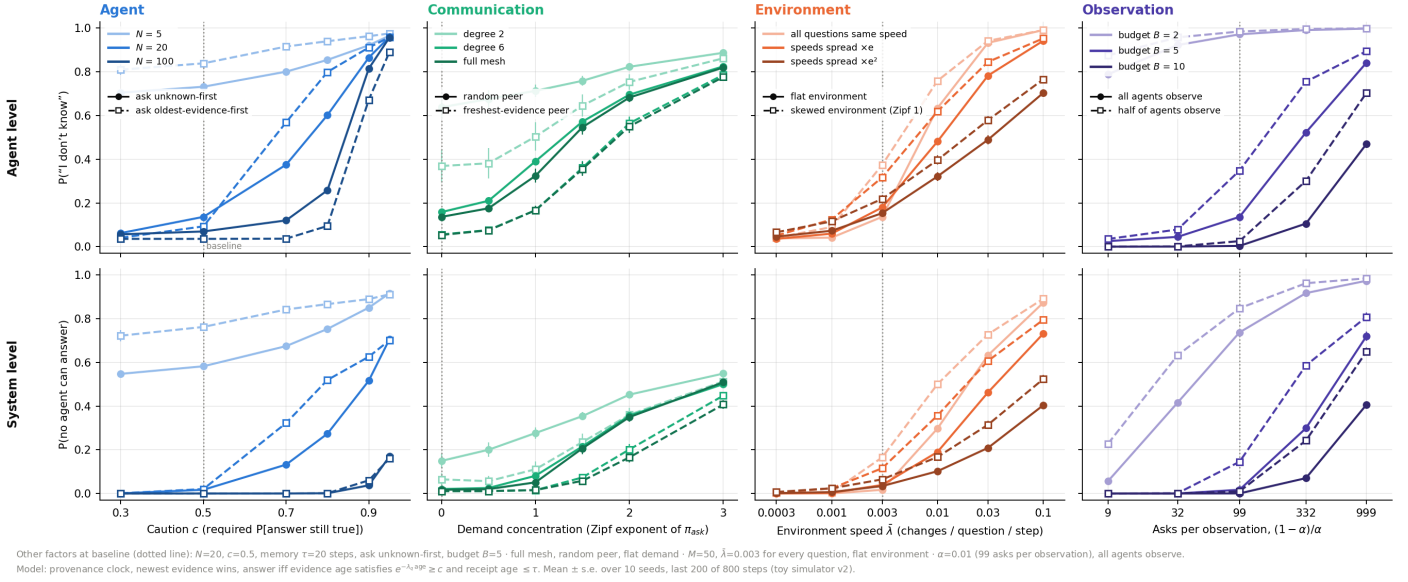


Figure 1: One intrinsic lever per factor carries the system from healthy to collapsed at both levels. Top row: agent-level $P(\text{IDK})$; bottom row: system-level $P(\text{no agent can answer})$; dotted vertical = the common baseline. Hue = factor; lightness = the factor’s second variable; solid-filled vs. dashed-hollow = its third.

function of $R_0 = K\tau_{\text{mem}}/M$ alone, with the transition at $R_0 \approx 1.25\text{--}1.5$ for $N \in \{5, 20, 100\}$ (slightly above the mean-field 1, from duplicate asks and founder extinctions); time to system collapse diverges approaching the threshold, and N sets how long extinction takes, not whether it happens. An environment channel decouples the two levels: at $R_0 = 0.5$, observation at $\varepsilon \geq 0.3$ keeps $\sim 95\%$ of questions alive somewhere while every agent remains $>90\%$ IDK — and the linearised SIS source term predicts the agent-level numbers to within a point (predicted 0.94, 0.80 at $\varepsilon = 0.3, 1$; observed 0.94, 0.81). Skewing which questions the world reveals is the environment’s real lever (IDK $0.03 \rightarrow 0.57$ at $M=100$ as normalised entropy falls to 0.2), and it matters precisely when the system leans on observation to stay alive. Skewing demand instead kills the tail invisibly: uniform-weighted IDK reaches 0.56 at Zipf $s=2$ while demand-weighted IDK sits at 0.01–0.04 — a probe that samples questions the way agents ask them cannot see the collapse. Collapse under rarity is a frontier moving inward, not a single threshold.

v2: every factor owns a lever to collapse (Fig. 1). *Agent — caution.* IDK rises from 0.06 at $c = 0.3$ to 0.96 at $c = 0.95$ (system death $0 \rightarrow 0.71$); small populations start collapsed ($N=5$: IDK 0.71 at $c = 0.3$, because observations per question scale with αBN). Asking oldest-evidence-first lowers agent-level IDK but raises system-level death (0.32 vs. 0.13 at $c = 0.7$): re-verifying what you hold spends asks that would have carried rare lineages. Raising caution trades stale for IDK (stale 0.19 $\rightarrow$ 0.00 as IDK $0.06 \rightarrow 0.96$); at baseline speed the not-correct optimum sits at or below $c = 0.3$. *Communication — demand concentration.* Zipf demand takes IDK $0.14 \rightarrow 0.82$ and death $0.02 \rightarrow 0.51$ ($s : 0 \rightarrow 3$) on the mesh; a degree-2 graph starts at IDK 0.64. Freshest-evidence peer selection roughly halves agent-level IDK at every s but delivers much less at the system level (death

0.41 vs. 0.51 at $s=3$): it redistributes evidence, it cannot create it. *Environment — speed.* Death goes $0.02 \rightarrow 0.87$ across $\bar{\lambda} = 0.003 \rightarrow 0.1$; through caution, speed is one of the steepest levers (in v1, with no caution, it had no effect on IDK at all). Spreading speeds across questions protects the mean (death 0.21 vs. 0.63 at $\bar{\lambda} = 0.03$, dispersion $\times e^2$): collapse concentrates on the volatile minority. *Observation — asks per observation.* From 99 to 999 asks per observation, death goes $0.02 \rightarrow 0.72$ at $B=5$; a budget of 10 tolerates roughly $3\times$ the ratio (death 0.41 at 999). The recurring structural finding is the agent/system split: policies and topology move agent-level ignorance freely, but only the observation channel creates evidence, and only memory-bandwidth-pool-size (R_0) decides whether created evidence survives.

Remaining tasks

1. Re-scope the planned experiment series to v2: the (m, τ, M) phase diagram inside the validity window, speed $\times$ caution (the closing feasibility window — headline), and topology/heterogeneity (who collapses first).
2. An analytic treatment of the feasibility window $\tau \in (M/m, \theta)$ and when it is empty.
3. Estimate the real LLM’s abstention function $P(\text{answer} | \text{evidence age}, \lambda)$ — caution c should be measured, not assumed.
4. Does the control chart see collapse coming? Slow collapse is a drift, and the paper shows adaptive charts miss slow drift.
5. Extend the repo simulator with the v2 variables for LLM validation of selected cells; blocked on an API key.